

DSA 8405 Probability and Stochastic Processes-20250925_173430UTC-Meeting Recording

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3h 6m 5s

 **Odhiambo, Collins** 4:00

OK. Good evening again. I'm going to start shortly. Give me two or so minutes, then we'll be starting.

So I think we can just move on from where we stopped yesterday.

So today, uh, we've just proceeded from where we stopped. Um.

So today I want to just do a recap of PGF so that it's a bit clear.

To all of us, I understand some of us we don't have.

Strong mathematical background or statistical background. So let me see what I can do so that we are up to speed.

OK, so just a recap of what happened yesterday.

We did look at. We tried to look at probability generating function, what we call PGF. Um.

Typically in probability in the theory of probability.

This course is called probability and stochastic process.

So stochastic is just another way of looking at data. So typically in the theory of probability, the most some of the most critical areas that you look at in probability is what we call random variables.

I should have done this at the beginning, but let me just look at it quickly. Then we have distributions.

So and then we have measurements.

OK, so in a typical probability class or after you have been taught probability from the start to the end.

To the maximum highest level of probability, you must be acquainted with what we call a random variable. So a random variable is basically a random measure. So random has to do with chance.

OK.

Random has to be the chance and then variable is changing is a measure that is changing that's that's the random variable. So it's so that the whole concept of random variable you need to have known what's a random by the end if you're doing probability if you're taking any.

Cause and probability.

Either from basic until you do advanced probability. So stochastic is like sort of an advanced version of probability. We have not even gone to the integration, the stellige integrations using probability.

I would be using stochastics, so which is those are quite advanced levels of probability. But at the end of it, the concept of probabilities, there's an idea about chance and how that chance can be represented numerical.

OK. And then there's the aspect of how the numerical data that is pulled out.

Um.

The numerical data that you have pulled out can be is distributed because that's very important how.

How the numerical data that represents the random variable?

Data that that is from the random variable because random variable is a measure.

So that so that that's how that numerical random variable after it it generates data like if I'm studying, let's say for example the age of students in this class.

So as I move from one person to the other.

Um, that's a random variable, so if I move from.

One person to the other, the variable is changing. So one person will be, let's say 28 years, another 131, another one 25. So as I move from one person, that's now what we call a random variable because that value is changing as you move from one person to the other.

So as you collect the variables, now you will have a set of data. Now the way the data is distributed, you may have more people at the middle of 2825, so a few people below. The way it's disputed is what now we call a distribution.

And that distribution is attached to some chance. It's attached to is associated with with chance. So that's now what we call a probability distribution.

OK, and that's why we write. We write it probability of random variable X is usually in capital letter, and that that random variable X could be age of students. The value it selects, it's usually in small letters.

OK, so so the probability that the event, the random variable X picks value 30. If I'm looking at the age of students, the probability that it picks value 30 is probably.

Certain value, let's say 0.3. OK, so that is the distribution. That's the idea of associating the random variable and the value speak to a probability distribution, like a function that maps the event to to a chance.

To a certain chance, tempor of chance that is given OK and of course that that has

some rules, so that that has come to some rules that the probability.

Um of that event lies between one and 0.

OK, so it has some rules and then the probability is non-negative.

Those are the some of the rules we have. OK, so it's is, is, is, is, is, is non negative.

And then if you sum, if you sum all probabilities.

So if you sum all probabilities, you should you should get one. So those are some of the rules that are there when you are defining that distribution. And then once you define the distribution, then you look at measurements.

That measurements.

So the measurement is is is critical for decision making.

So this measurement is critical for for for decision making.

So the measurements will involve the mean measures of central tendency to involve dispersion measures of dispersion like standard deviation or various.

OK, can involve skewness.

Just to show how the data is skewed, it it it it does involve kurtosis.

So this these are measurements that now will help you make decisions regarding the random variable. So because at the end of it you want to make decisions.

OK, so at the end of it, uh.

You want to make option. You want to at the end of it. You want to make options.

You want to make a decision.

You don't just want to model the data, look at the data. You just want to make some decisions.

Right.

So that that's the basics of the whole idea of probability.

I can now delve into other aspects of probability. Random variable can either be continuous on and continuous or discrete. So it can either be discrete or continuous.

So if a random variable is discrete, then we talk of a probability distribution, a discrete probability distribution.

If a random variable is continuous, then we talk of probability density function and that involve a bit of calculus, integral calculus for you to compute the measurements.

OK, when it is discrete, it's still a bit of algebra. It's mostly it will go to the world of algebra. If if your distribution is discrete, there will be a bit of calculus, but mostly usually it goes to the world of algebra and sometimes.

Um, yeah, it has to go to algebra for you to now generate measures.

OK. And of course measurements are good for decision making. The reason why you

do all that is because you want to make decisions. Then at the end of it, you want to make decisions. So once you know the mean, you know the variance, you know the skewness, kurtosis.

You have other measurements like interquartile range if you want to look at the median. So you have all these.

Uh, aspects.

Yeah. So you have, you have to make, you have to look at the range, you have to look at other aspects that that that may come.

Right, so.

So if if the this type of random variable here takes the stochastic format.

Then eventually when you start making decision, you want to derive measurements so that you can make decisions regarding your data. Some approaches techniques are important.

OK, some techniques for deriving the measurements are very important because after.

After you've determined that your data is probably a pure, pure birth and death process. So looking at this characteristic of data, it is pure birth and death process.

Uh um.

So it's it's pure bad and pure death process.

Once you've determined that and then.

Uh.

So once you've determined that the structure of your data which may pass through the distribution, then when it comes to making determining the measurements, you may find difficulties here. So that's why there are several techniques that are available in the theory of probability.

Uh, that are helpful to to to that are helpful.

OK.

OK, so let me just revise a very simple case, like what we did yesterday. So I'll not take the whole topic of probabilities long. It's huge and I think you may have covered pieces here and there once in a while.

And here I'm talking when I'm talking of pieces, I'm talking of.

I'm talking of a situation where you are looking at it from a distribution point of view, not not the high school kind of probability, but you're looking at it from a distribution point of view.

So yesterday we looked at a simple, a very simple case because I just want to focus

on probability generating functions as a technique or as a tool that is important. When you want to compute measurements from our probability distribution or from our probability structure. And so this provides very easy unique tool that can be helpful because each.

Because each probability distribution has a unique probability generating function the same way each each probability.

Distribution function is unique is unique to to specific random variables.

So each probability distribution is unique to like if we are talking of counting process, the arrival date. Once we describe the whole arrival process and it's Poisson, then it has a Poisson distribution. That Poisson distribution is unique, is a unique distribution. It's well defined.

Unique probability distribution. Also some of these facilities like moment generating function which we will not cover.

Uh.

This one will not come.

MDF.

And another facility like probability which we will cover.

OK.

PGF.

So this facilities are also unique. So a Poisson, a random variable that follows Poisson has a unique moment generating function.

OK, a random variable that follows exponential follows Weibull, Cauchy, all these type of distributions that are there. Normal distribution, gamma distribution, all those distributions that are there there.

Their MGF if it exists. If the MGF exists, then it's unique to that distribution.

The probability generating function, if it exists, then it is unique to that distribution.

The only difference between these two is this is applicable to all the variables. OK, so this is applicable.

To to all variables, all all random variables and we introduce what we call the expectation of XT as a function we introduce into.

Into the distribution.

Um.

Looks like I disappeared.

Try again.

OK, so the difference between the PGF and the MGF, the PGF is applicable to all

random variables, whether continuous, whether discrete. And then for PGF, for MGF we do the expectation of E^{sX} .

OK, on the other hand, the probability generating function, the PGF is only applicable to discrete.

So it's only applicable to discrete probability distributions.

It's only applicable to discrete probability distribution and for PGF we do we do the expectation of S .

X .

So we do the expectation. In other words, we introduce S power X and then you get to the summation for that period. So a good example is a good example. Let's start with a very simple example.

Uh, like, uh, Bernoulli.

So like a Bernoulli distribution, this is a very simple a Bernoulli distribution. It's defined as correct. It should be. It should be P .

Really.

One minute of X .

So X can take value.

It can take value 0.

On one.

0 L so.

OK, so X can take value 0. So this is how a Bernoulli. So very simple and of course $P + Q = 1$, so is a dichotomous outcome in the success of A . Probability of success is P , probability of A is Q .

If you add $P + Q$ you should get one and then the distribution is defined this way. $P^X Q^{1-X}$, X and X can only take two values, value 0 and value 1. So if you want to get the.

Mean the measure. So as you go, you define the probability. The next thing is to get the measurements, want to get the measure, which is the mean and the variance of this distribution using the probability generating function.

So first of all we we define the probability generating function PS .

So PS is defined as the sum. Of course, this is being a Bernoulli. A Bernoulli runs between zero and one by definition of a Bernoulli distribution. So you introduce SX . And then now you you bring in the distribution.

So you introduce S of AS power X and then you bring in the distribution and then you sound it over that period.

So if you try to unpack this, if you unpack this, that means that you start with 0 where there's X you put zero, you'll have $S0P0Q$.

One man has a leader.

Trying to unpack that plus now you go to 1, so X can X can only take zero and one.

OK.

Because X can only take value zero and one. So if you do that, anything power zero is 1, anything power zero is 1. This remains Q. So here you'll have Q plus SB.

KQ plus SP anything power zero is 1.

So that becomes the PGF of a Bernoulli distribution. That becomes the PGF of a Bernoulli distribution. So to get the mean and the variance, first of all to get the mean as I remember the way we said the mean.

You just differentiate with respect to S and then replace S with one after differentiation. So if we do so the mean.

So you differentiate with respect to S, you differentiate the probability, the PGF of that Bernoulli.

So the derivative we have with respect to S, this will be 0, this will be.

So there's no S, so you don't actually need to change it. So the answer is P&P is a mean. If it was PS or something S, then we'd have to replace S with one.

So the P becomes the meaning of a Bernoulli distribution.

Right.

If I want to get the variance.

Remember the variance was you do the DDS twice.

Yeah.

Then then you will replace that with one, then you add ES, then you subtract. So you add the mean whatever you got here, the DES.

Of PS.

And then you subtract now the the DDS of PDS. Now is we subtract the square. OK, so you so you differentiate twice.

You differentiate, differentiate twice and then replace S to be one. Then you add the first differentiation, then replace S to be one. Then you subtract by the square so that.

So that if you look at so if this is the first differentiation, this one here is the first differentiate is P So the second one there is no S so the second one becomes 0 plus the first one that is still P.

Minus the second, the first one squared, so it'd be.

OK.

So that's the.

Um.

That's the variance of a Bernoulli, a very simple Bernoulli distribution.

I think I can do a binomial and then.

After that I can do. Now we can go back so that we see how the probability distribution probability generating function works. So this is just a basic one for Bernoulli. What if we try to?

Look to a more. Let's look at uh binomial.

It's got binomial. It's a popular distribution.

Or a geometric. Let's just look at binomial. So binomial see I hope I'm right. The probability for a binomial if if X follows a binomial distribution.

Then the the probability distribution function is should be NX .

PX^{QN} minus X and then X can go to 01 up to N .

0 answer.

OK.

So that can go up to end.

0 elsewhere. So that's the probability distribution for a binomial distribution. So a binomial is just an extension of a Bernoulli where now you have.

You have samples up to N . Remember for by Bernoulli it was only zero and one.

But now for binomial is up to N .

So the the coefficients are the possible combinations. So it's success or fail. P is success, Q is fail. Probability of success is P The probability of fail is QX is a random variable.

This is a combination and combination X or possible combinations of success and fail and this is very much applicable in quality control area.

Right. Um, so if that's.

If if that's the case of the case, then I can do. So if that's the case, so let's use the PGF to get the mean and the variance of binomial distribution. So the PGF.

The probability generating function PS the PGF you just sum. So the definition of a binomial is X moves from zero to M .

We introduce S of X .

And then we have N combination X . Yeah, we have PX . Yeah, we have QN .

So we tried to solve this binomial. So this is a PGF of.

EGF.

OK so so to solve this. So the best way to solve this is I would see combine the most combine the ones that share common things. So I can do I can do it this way. So this should be this not it's not as it's not it's. It is defined as N in a binomial, so it's. So I can just rearrange my work here.

SP.

Yeah, yeah, it's 'cause s s as a power X and it's P as power X or.

Yeah, so who can tell me what is this?

In terms of uh 'cause that's expanded form, but in terms of.

After factorial form, what would be that?

All right, let me take you first then to some few things before we can come here. It's just a bit of algebra so you can see what I'm trying to see.

What is this $A + B$?

What is $a + b^2$? Can somebody tell me what is this?

wo **Walter Ouma** 36:59

A squared plus.

sk **Stephen Korir** 36:59

Is it $S^2 + 2AB$ plus B^2 ? Oh, OK, well time.

oc **Odhiambo, Collins** 37:05

Thanks. Uh, what about this?

Well done. What about that?

wo **Walter Ouma** 37:22

A cubed.

oc **Odhiambo, Collins** 37:26

Thank you guys.

wo **Walter Ouma** 37:29

Last, uh, that will be two $A^2 B$.

oc **Odhiambo, Collins** 37:35

It should be 3.

wo **Walter Ouma** 37:36

3.

oc **Odhiambo, Collins** 37:37

A square B plus.

wo **Walter Ouma** 37:42

Plus three AB squared.

oc **Odhiambo, Collins** 37:43

Yeah.

Yes.

wo **Walter Ouma** 37:48

That's the big cute.

oc **Odhiambo, Collins** 37:48

That's big key. Thanks, thanks, thanks, thanks. So there's there's a pattern here that Walter is following. Let me try something like this.

There's a pattern. Volta is following it. What will be that?

This this will be. There's a pattern he's he's trying to follow. This will be a power 5B power 0.

wo **Walter Ouma** 38:17

It's about 5.

oc **Odhiambo, Collins** 38:23

Plus you see the power of A keeps on reducing. First of all, let's let's put the powers first. So this will be a power 4B power one. This will be a power 3B.

To this will be.

wo **Walter Ouma** 38:42

It.

OC **Odhiambo, Collins** 38:42

A power 2B power power 3.

OK, this would be a power 1B and then this would be a power 0B.

WO **Walter Ouma** 38:48

Yes.

OC **Odhiambo, Collins** 38:58

So you notice that the power of. So this it's like you bring these factors A&B. You start with the maximum power for one. So one goes down 5 for up to 0, but the other one moves up from power zero going up way.

Upward and then the coefficients are the combinations.

Coefficients are the.

Combinations. I think this is just basic algebra. I don't know if they're in high school or.

They call it what? Is it the binomial expansion? Then this 5⁴ and this is 5/5.

OK, so that so you start with, you start with one factor, the power goes on reducing while the other one the power goes on increasing, then the coefficients are the combinations, so the combinations.

So like if I say 3 combination 2, what what this means is just three factorial, 2 factorial and one factorial and and three factorial is just $3 * 3 * 3 * 2 * 1$.

You can go and reduce it.

So.

So that here 5 uh factorial.

05 combination 0 is actually one because 55 combination 0 that's like 5.

Factoria 5 factorial is 0 factorial is 1.

So here you'll end up with and anything power zero is 1, anything power zero is 1. So you'll end up with a 5 + 5 A 4 plus what's this that that's like 5 factorial.

WO **Walter Ouma** 40:54

OK.

2.

OC **Odhiambo, Collins** 41:06

3 factorial, 2 factorial. That's like $5 * 4 * 3$ now 2.

So this should be 10. Is it 10?

$5 * 4 * 3$ so this should be 10.

10 A³B² plus 10A²B³ plus 5AB⁴ plus B⁵.

All right, so.

That's that's what these are just algebra for expansion. So if you're looking at if you have a + b or something, then you do expansion. So you'll have what this what we call a binomial expansion.

Wo **Walter Ouma** 42:01

Mhm.

oc **Odhiambo, Collins** 42:02

So this is what we call a binomial. So if you have, if you have.

Wo **Walter Ouma** 42:06

It.

oc **Odhiambo, Collins** 42:09

If you have a factor written in this way, power something you can generalize using binomial expansion.

Because it follows certain patterns now.

This expanded form, we can also rewrite it. You can summarize this as a summation, so we can summarize that as a summation of l starting from zero to 5.

Then we have $5l$.

And then we have, uh, beam.

$lK 5 - l$.

OK, So what I've done here, I've tried to summarize this, the whole of this part, see the whole of that part.

But.

I've tried to summarize it in summation because if you open up the summation, if you unpack the summation, you start with 0, so everywhere there is l put zero. So if you put 0 here, 0 here, 0 here, you'll end up with this.

Then you put 11 where there's l , 11, you'll end up with that. Then if you put where there's l is too, you end up. So what I'm saying is that the whole of this expression

here.

OK, so the whole of that expression there can be summarized in this form.

You can summarize it in that form. So that means then if I tell you $a + b$ power 10, I can summarize that as a summation of let's say I or X can give it value X.

And.

10 power X.

Any of these, the order doesn't matter.

They are committed.

OK, so if I give you that telly to expand, then I can summarize the expansion as a summation.

As a summation of that where it is $a * B$, the value of the power of one value keeps on reducing while the power of the other one increases and then the coefficients are the possible combinations.

Now if we come back here at the PGF, the probability generating function. Yeah, if you look at this PGF here.

So the PGF.

of a binomial distribution.

Is we had already established up here.

That is giving us that.

So it's this sum our N.

Our N moves, our X moves from zero to north and then I have NX.

Then I have ES or SQ.

Then I have Q.

And.

OK.

So that's a PGF. Now the idea of solving this is now can we rewrite this in this form?

Let me write it in that form.

So if you're given.

The factorized version and we are told to expand. You can write it in this expanded form.

But can you? Can you do the opposite? Like can you?

Um.

Can you can you do the reverse? If you're given the expanded form, can you factorize? Yes, so that's the ABM. So this one here then this is same as ES plus Q.

Or N So if you write it in expanded form, that's what you.

It's like you're given already expanded from your talk to factorize it. So if you're if you're able to go forward, can you go backward?

We can move forward. Is there a way you can can move?

It.

Any question?

OK, so so if that's if that's the PGF, then to get the mean, so that's a PGF.

Of a binomial distribution, so together mean.

Oh.

We just need to do the derivative and then after that we replace S and one.

We just get the derivative, so we come here. So now we get the derivative with respect to S.

Yes.

So then everybody with respect to S Now this this will take you to chain rule because there's a function inside a function. So the outer function is you put N outside so that now the inner function remains as point.

Um.

So you start with the outer.

Using chain rule, sorry the outer function. So you bring the N on this side, then you reduce the power of N by one and then you multiply it by the the inner function, the derivative of the inner function. So if you differentiate this one with respect to to S inside here.

You only get pee.

Because it's with respect to S OK and then so that's the first derivative. So for us to get the mean we we replace S with one. So if I take put S to be 1 here.

It will be 1P plus QP plus Q is 1. Anything power one is 1, so the answer should be N.

So the answer should be and and peak.

OK.

So if I need to get the variance, the variance is made-up of the variance of X is made-up of.

You have to differentiate twice.

With respect to S.

And then you add the first derivative with respect to S.

And then you subtract.

The first delivery team with respect to SES square the other one.

So that's how you get the lines. So you differentiate twice and then you add the one

that you had, then you go again and subtract the square. So if you if you if we if we differentiate twice here.

So if you take the second derivative, so the first derivative here is this. This is the first derivative.

OK, so that's the first derivative. Now the 2nd derivative is now differentiated again.

So if you do the 2nd derivative there, I will have $N - 1$.

And.

And -2 .

And then multiplied by the internal whatever I use so it's PVS that should be 2 square.

Yeah.

Plus.

Now you will have to to replace S with one. So this one of course we know it when you replace S with one we got NP .

This one we know it only replaces with the one you get a piece or just quite.

Yes, right.

So replacing S with one inside here I'll have if you replace S with one here, your PP plus Q is 1. Anything power one is.

So this side here will give me $N - 1$ N .

V square.

Plus M minus M square.

So this part here will give me N squared P squared minus NP squared.

So it's just algebra anyway, the rest is just algebra, so.

So that becomes the variance. So we've found the mean as NP , the variance as NPQ .

So what we have done here is just to establish the mean and variance using the probability generating function.

Technique. So using the PGF technique. So remember first of all you define the PGF, which is the expectation of S power X . Once you solve it, then you get the mean.

Then you get a the variance.

Yeah.

Right. Let me give you a break of 10 minutes, then we come and continue to see the application of this in death and death process.

OK, so welcome back. Um.

I think after going through that.

It should be easy now to see, uh, the other parts.

Of birth and death processes.

Right. So just a recap of what we've covered yesterday is that the characteristic of a pure birth process, which we call Yule-Furry process.

Has the following, so so you consider a cell that follows the following rules. So a cell present at time T has the probability of.

λH So that λH can be $\lambda \Delta T$ So if you look at what we used here is more ΔT , so that's similar as H So instead of H we just use ΔT here.

Yeah, ΔT here. So it's the same thing. It means the same thing as H Different books use different approaches.

So the probability then is independent. So I think we did cover this, but.

So we approached it non using non probabilistic approach. OK, so if N_T is the number of cells at time T then if you add the small change that H or that small change.

ΔT is the same as the amount being up to that time and then plus the small change. So if we divide across by the small change and introduce limits, then we have.

Differentiation from first principle. So if you have this case, this is the same as $\frac{d}{dt}$ of $\log$ and we did so that we saw that yesterday and of course doing the integration. So if integrate both sides this will be.

If we if we bring this $\lambda \Delta T$ here, it will be integrated to be ΔT plus a constant. So that constant is now what we call N_0 at time 0. So we looked at that. We derived that yesterday using non probabilistic approach.

The same can be applied if you look up at a probabilistic approach where we were looking at if the arrival is this. So that means if there's no arrival, we break it the way we did for Poisson.

There's a way. There's a way we we did it for for Poisson.

So until you come up with this type of equation.

And then you put the initial condition to be one cell and then of course we solve. If you solve this condition with this condition, you'll end up with that, which is.

Still what we call a non negative, what we call a negative binomial distribution.

So I do want to go back there, or rather to bring the technicality of of a negative binomial.

But there's something about the probability generating function.

So probably we look at this geometric because this one works well, especially for most most of the cases we are looking at pure birth. If we look at the branching

process, there will be an aspect of geometric.

So this is how a geometric is defined and therefore if we define geometric as this, let me see if we can continue here a little bit with our white board.

Just to make sure that we've captured that section of probability density generating function. So this this is a case of a binomial distribution. Binomial distribution is largely applicable in quality control.

In marketing or where you have a binary outcome and you have, we have sampled some few cases. In quality control you'll talk of defective items versus non defective items and then you sample a certain.

Number of items.

So the geometric is different.

They are both extensions of a Bernoulli distribution, but the way we look at geometric distribution.

Is we are looking at to success until you have first success. So you have fail, fail, fail, fail until you have the first success. So the distribution, that type of distribution is what follows a geometric distribution.

Because you have failed, fail, fail until you get one success. OK, so for example, if you are, let's say you're modelling quality control in a lab, in a lab, in a lab that is testing.

Or let me just think of quality control in a manufacturing company. So you have machines in a manufacturing company that are working 24 hours generating products. So you put there are several inputs or raw materials you put.

And that machine works and then generates a soap or let's let's talk of a toothpaste like hold it. OK, so there are several raw materials that are put into the machine at certain quantities.

And um, under certain pressure and under certain.

Atmospheric pressure and under certain heat, so they generate on the toothpaste or just just look at think of crude oil, crude oil.

Crude oil in its state cannot be helpful until it is processed, and processing means that it is subjected under under certain pressure and and temperature. So you manipulate temperature and pressure.

At certain level, then it passes through certain machines to as you manipulate temperature and pressure so that you will end up getting what you need, like a gas, petrol, petroleum, whatever. So those are products of crude oil.

Now that that machine that is running, it's running to it for hours.

Now they are.

There are quality control people who will sample few of the project products over time and to ensure that the machine functions at a more efficient and it's also effective in its work.

So there will be sampling products after a while. So if their first product is that is not defective, it's non defective, that's fine. So you sample some until you get the product that is defective so so.

Then you put that product aside, then you sample, sample sample until you get a product that is defective. So that kind of a distribution is a geometric where you have fail, fail, fail until you have success and success here can be looked as.

A defective item so that you can, once you get the distribution and compute the measure, then you can recommend the machine to be either repaired or stopped, checked or something like that.

So the the geometric distribution is where you have that. So the distribution is actually fast success and then the queues are the queues are there to.

So you have Q^k up to the first success, where X can be one to to Infinity.

So this is this is what what we call a geometric.

Distribution.

OK, so the geometric distribution is defined as not zero elsewhere.

So you have fail, fail, fail until you get success. Fail, fail, fail until you get success. OK, so getting the PGF over geometric progression.

Uh, so the PS of a geometric location use some. So remember this one now moves from 1:00.

To Infinity.

To introduce S power X , then you have $PQ^X - 1$.

Right, so we tried to solve that this this one.

I'm just adjust it to make life be easier here.

So I can do, I can pull out SP , I can pull out S .

That I have.

One it still remains so S because I pulled out.

SP there, so I have that.

Schools now that.

So this will be SP sum.

It's equals to 1.

SQ .

Minus one.

OK, so that that would be SQ^{-1} . So if you try to unpack this, so I have SP here.

If I try to unpack this, just open it up, start with one. So you'll have SQ.

0 to start with one.

SK1.

Excuse to.

That way to Infinity.

Right, so this this is the same as SP.

One plus SQ.

SQ.

Mhm.

So that's up to Infinity. So this is a geometric progression.

That's a geometric progression. So and the sum of a geometric progression to Infinity, we have seen that that sum is actually $1 - R$ where R is the common ratio. So that's just this.

So then you are probably generating function is SP.

All over one.

Let's keep.

Right. So that's your PGF. This is your.

This is a PDF.

OK, so that's your PDF.

So to get the mean and the variance, you do the same thing we've done. So the mean just the.

Uh, you get the DDS of the PGF.

And then you so you so you get the the DDS of the PGF and then of course you give then you replace SU one.

And then the variance you can do the same. So I don't want to go there for that part, but I can leave it there for you to look at it because this one this is a quotient will follow a quotient because it's the derivative of quotient. So you follow what we call quotient rule.

But I wanted to show us that eventually you will derive. When you derive the pitch here for geometric, you will end up with that. I think the rest are more algebra there.

I'll leave it for you to try to work out, get the mean.

And the variance using the PGF.

So for for mean you have to differentiate this with respect to S and then after getting the value differentiate you replace S to be one. For the variance you have to

differentiate twice.

So you have to differentiate twice.

And then you replace S to be one, then you add the mean.

What we got as the mean, then subtract the mean.

Square. So what you got here will meet here when you subtract and then you have some relay variance.

PG.

OK.

Uh, this guy here used the MGF this and they didn't use it.

It's MGF that you introduce the E power, so that's not a good one.

What's the MGF?

Right. So that's a P_4 pure bad process. Um.

Poison. We did look at this, so we're not going to that.

So now when we consider birth and death, I'll just finish this and then we'll stop there.

So when we consider now birth and death together, you consider the transitions from E .

NN to $N - 1$ as well as N to N plus that N greater than one.

So we so we only allow the transition if we are moving from zero to 1. So assume the process is at T is this. So moving from N to $n + 1$, this is the probability.

But now moving from N to $n - 1$, we have this. So it's like you want to combine if there is a bath. You see if you add one, that means there's a bath, a new bath. The probability is this. But if you move this, that means there's one death.

So the probability of death is this.

OK, so that's how we reason for birth and death process. So if you look at those Kolmogorov forward equations.

If you look at the Kolmogorov forward equation, so moving from.

And if there is a value, if there is one, so you can either have bath or they can either have death so.

So this is how you reason. So if there's any increase, then we have a bath process. If there is a decrease, then we have a death process.

So the probability, so the probability that OK, so come here if you look at the probability that they are N .

Entities in that system at point $T + 1$ is the same as.

When there was, by the time you are at this point, you already had reached the value.

OK, so there was no death and there was no birth. Or in the previous setting there was one birth, or in the forward setting there was death.

Yeah.

OK, so let me just repeat it again so that the Kolmogorov equations, if you remember how we were using Kolmogorov equation for Poisson at time T , the probability that there are N entities at time.

$P + H$ Remember H is a small change, the probability that there are N entities at that time.

Plus that small change, it means that probably the first situation is that within that small change H there was no birth and there was no death. So we are we were at N .

So there are two scenarios. There are three scenarios either at that point of at at that point when there was slight increase. So let me just see if I can demonstrate it well on why probably it will make more sense.

For a birth and death process. So I have so this this stochastic system, this is a stochastic process. So we started at $.0$ so we have at $.0$ point 1.2 .

$.3$ that way up to point T .

OK, so the number of people and the number of cells all through.

If you include those who die or those who who new new people coming, so the number of cells at this point and then so this PNT is a probability that.

At that point T you have N cells.

So N cells. So it's a probability that you have N cells at at point at point T .

OK, so you have answers at one T So if you include include HH here, H is a in between here. So the the small change here is what calling H small change so.

So if if I'm looking at PNT plus that small change, it means that if I'm looking at that probability, it means that there are three scenarios #1.

After that small change, I have hand tea and therefore there was no there was no increase.

Plus minus then there's no there's no death.

Or.

At PN previous time there was a new there's one new path or.

At at time PN plus one, there's one exit.

So that's that's how we interpret it. That's how we interpret it.

So that uh.

So that if you reorganize and then bring the like terms together. So if you open the

bracket, reorganize the way we did and then bring the like terms together, bring the P this side and then introduce divide everywhere by H.

And then to reach the limits, then you'll have this kind of a scenario.

And you'll have this kind of. This is just simple algebra, like you open up this P here, P everywhere, then you bring the like terms together and then you bring this PT on this side.

And then you divide everywhere by H, then introduce the limits. That will give you a differentiation, so you'll end up with this kind of situation.

So for situation where you have zero, if you replace with 0 you have that. So you assume that the first initial condition is 0 is 1.

And then now you can start using iterations to get it. So you can start with when P is 1, this is what you got. P0 gives you that when P so you end up.

Getting that until the initial condition disappear.

Right.

OK, so.

So it's like you're putting T here. So if you put one, put one, there are 111111.

Assuming that if you put one here, you'll end up with with that if you put and then you replace zero with 0.

Uh P1 so this one can be replaced. Therefore P can become that.

Uh, so if you put P is one, you end up with this.

If you put PS2, you end up with that pathway. Um.

So this is where now you can relate unless until you reach that that type of that type which will now give you PN is the product of all that until you get P1P0 and all that until you get to MU N.

You can also look at it in terms of Markov chains as transitions.

Can also look at it. I post this if probably you're interested with that and you can also look at it from a linear birth and death process.

Right. Uh, home.

So that's of the aspect of a birth and death process. I don't want to go more than there, so that I don't want to introduce more than computations. So let me stop there today.

OC **Odhiambo, Collins** 1:31:07

We are coming on. We should be done. I think we should be done by next next week once we are done, once we do a bit of martingales, then the other week we can revise and then I think we should be done, yeah.

CK **Cosmas Kungu** 1:31:25

OK. And thank you for this kind of arrangement and my request is to that this PDF, the one we are showing us here, you upload it on the Google.

OC **Odhiambo, Collins** 1:31:25

Yes, yes, yes.

CK **Cosmas Kungu** 1:31:42

Classroom. I note it was not part of the Google stats.

OC **Odhiambo, Collins** 1:31:47

OK.

CK **Cosmas Kungu** 1:31:49

And also those other notes you have been scribbling on there.

OC **Odhiambo, Collins** 1:31:53

Oh, yeah.

CK **Cosmas Kungu** 1:31:54

On there, if you are able to, it will really help. Right now we are trying to escape with the lightest basket. Yeah, we are juggling. So yeah, thank you.

OC **Odhiambo, Collins** 1:32:03

OK.

I I understand. I I actually want to also put the questions as early as possible. I think in the course of the week the question for that can you can help you in exams like to practice.

All variety of questions that can that will come out of this course. So that will also I'll I'll send it as early. I'll post it as early this week.

CK **Cosmas Kungu** 1:32:35

Thank you. Thank you.

OC **Odhiambo, Collins** 1:32:39

I know stochastic is usually it's usually a bit advanced anyway, so I I've also taught students here. I think at UCLA I used to teach an advanced course in in stochastic process.

And it's it's it's it's not unusual for people to struggle to understand it's it's not unusual. So it's it's quite usual for people not to struggle. It takes it takes a bit of time especially.

Uh.

Especially.

If if it's an area of your interest, it will take it will need you to take a bit of for me. I can tell you for free. It took me some weeks to just study stochastic and that time I was in third year in my first when I was doing my first degree.

To me some way, I was not reading anything. I was just sat down to understand the concept of stochastic. It's it's a bit advanced. So that's what I can tell you. So partly because.

Of of utilizing different advanced applications in mathematics and applied mathematics.

It's like somebody was it's requires somebody who has already done ordinary differential equation and you know the is taught at fourth year for those people who are doing who are doing double maths or undergraduate who are doing mathematics.

On the differential questions are usually taught in your either 3rd or 4th year after you've already done, after you've been grounded in calculates, after you're good in calculus. Now then we assume you're good in calculus. Now we can teach.

And I wasn't taught a DE in school because I did statistics, but I had to read on my own and and so that I can understand the concept of stochastic. So I I know, I know there's a there's a master's course.

I think I taught it shortly in Illinois, Illinois Tech. It's a master's course in data science and one of the course I was teaching was stochastic. So students were like.

How can we avoid all these things? Is there a way we can move around and avoid all these things? It's just that if you are as a degree program, you're also required to have a bit of grounding.

In stochastic, in the event that some of you will come later to do PhD, OK, because the your degree is not just to help you execute your functions in your work as a. As a data analyst, part of the degree of the master's programme is also to prepare you for PhD. So that's why you can see things like stochastic are there to prepare you because in practice, if you're it's just for industry work, you may not actually need most of these things.

For industry work, but for if you're doing it for advanced knowledge where you want to advance some aspects in your PhD, you'll need a bit of grounding in some of these areas. OK, so that's that's usually the reason.

Why some master's degree? You cannot completely have everything analytics. You still need some bits of theory and and it's not unusual if you don't understand at first. It's normal. It takes some of them takes time. You have to read several times.

Um, for you to become a mastery in the concepts themselves.

OK.

But don't worry, exam will not be like scratching as such because you are a data scientist. It's exam will be moderate, but I'll I'll send you the I'll I'll post the questions that can help you prepare.

All right, guys, so let's meet next week. I think we have very little in.

We have some few areas in.

In Martingale's theory of Martingale's and then branching processes, some various areas in stochastic processes, touching here and there, and then I think we should be done.

Alright, enjoy the rest of your evening.

SK **Stephen Korir** 1:37:35

Thank you.

SO **Sharon Owino** 1:37:37

Thank you.

GN **Graham Nyasoko** 1:37:39

Thank you, bro.

☒ stopped transcription